



The Autonomous Sovereign AI Operating System for Paytm Merchants

Transforming over 10 million Paytm Soundboxes from passive payment speakers into proactive, multilingual business partners.

Qora continuously audits supplier invoices, reclaims lost credit, hedges cashflow deficits, and executes autonomous restocking in 23 native Indian languages.

SARVAM AI

Voice Copilot

Saaras v4 STT + Bulbul TTS + Mayura translation across 23 languages with colloquial code-mixing.

COGNEE

Memory Graph

E-C-L pipeline creating a persistent knowledge graph of customer habits, pricing contracts, and seasonality.

n8n

Agentic Workflows

Event-driven orchestrations automating invoice OCR verification, WhatsApp reminders, and PO dispatches.

PAYTM CORE

Sachet Capital

Soundbox transaction velocity feeds instant micro-credit underwriting with daily automated DBR sweeps.

India's 30 Million Merchants Face a Silent Margin Crisis

While digital payments have been solved by Paytm Soundbox, the core back-office of Indian retail still runs on mental arithmetic, crumpled paper receipts, and unrecorded credit.

5% – 14%

Supplier Rate Inflation

Distributors routinely creep unit prices on fast-moving SKUs. Kirana owners purchasing 150+ items cannot verify contracted rates across dozens of weekly invoices.

ANNUAL ESTIMATED LEAKAGE: ₹38,000+ PER STORE

22% of GMV

Trapped Khata Credit

Informal customer credit (khata) is essential for customer retention but suffers high default rates. Merchants avoid calling regular customers due to social friction.

68% REPORT UNRECOVERED CUSTOMER DUES

4.5 Hours

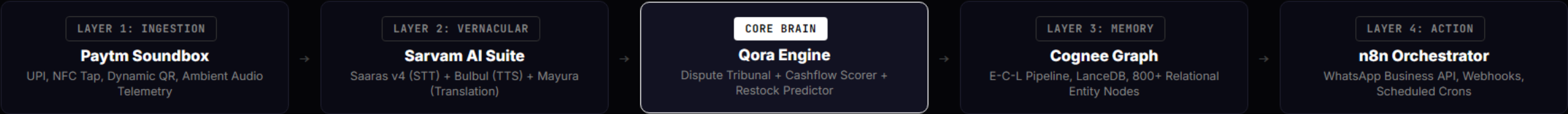
Daily Operational Friction

Manual ledger reconciliation, stock counting, supplier phone coordination, and cashflow balancing drain energy from business expansion and customer service.

NO FORMAL BALANCE SHEET TO QUALIFY FOR BANK CREDIT

The Strategic Opportunity: The merchant already trusts the Paytm Soundbox on their counter. By infusing it with sovereign voice AI and continuous memory, we convert an audio alert box into a proactive Chief Operating Officer.

The 4-Layer Cognitive Merchant Architecture



COMPONENT	UNDERLYING TECHNOLOGY	EXACT FUNCTIONAL RESPONSIBILITY
Voice Interface	Sarvam Saaras v4 + Bulbul TTS	Real-time streaming speech recognition in 23 languages. Native support for code-mixed Indian speech (Hinglish, Tanglish) with zero UI touch requirements.
Document Intelligence	Sarvam Vision (3B SSM) + n8n OCR	Instant parsing of WhatsApp photo invoices, extracting complex Indian handwritten & printed tax invoices with itemized rate verification.
Agentic Memory	Cognee Graph RAG + LanceDB	Maintains long-term semantic state across customer payment habits, distributor price histories, and seasonal demand fluctuations.
Automation Workflows	n8n Event Engine	Triggers autonomous WhatsApp payment reminders, drafts supplier purchase orders, and delivers daily executive P&L voice summaries at 10 PM.

Engine Innovations That Put Qora Ahead

INNOVATION 01

Federated Regional Price Oracle

Single-store audit only catches rate changes against past bills. Qora anonymously pools invoice data across neighborhood stores within the same postal code.

- Hashed zero-knowledge price and SKU extraction per distributor GSTIN.
- Detects regional price discrimination: flags when a distributor charges Store A ₹112 while neighborhood peers pay ₹104.
- Arms small retailers with collective bargaining leverage against monopolistic distributors.

INNOVATION 02

Ambient Acoustic Lost-Sale Telemetry

Zero-touch demand tracking via on-device acoustic keyword spotting on the Soundbox hardware.

- Privacy-first on-device phoneme detection: identifies merchant phrases like "*Nahi bhaiyya, khatam ho gaya*" (Out of stock).
- Captures unmet consumer demand in real time without requiring manual stock counts or screen typing.
- Logs lost revenue directly into Cognee's replenishment graph to adjust future reorder thresholds.

INNOVATION 03

Autonomous Multi-Agent Dispute Tribunal

Takes the pain out of supplier confrontations by deploying autonomous agent negotiations.

- Auditor Agent flags invoice anomalies against contracted rates and purchase orders.
- Evidence Builder Agent compiles delivery receipts, PO timestamps, and rate contracts.
- Negotiation Agent dispatches automated, professional WhatsApp claims to distributor billing desks to issue credit notes.

INNOVATION 04

Dynamic Perishable Markdown Engine

Prevents total margin write-offs on perishable stock (dairy, bread, produce) nearing expiry.

- Tracks shelf-life decay clocks initiated on stock check-in.
- Calculates burn velocity vs. remaining hours to store closing.
- Triggers Soundbox voice announcements or WhatsApp broadcast deals for nearby regular customers to clear inventory at break-even.

Automated Working Capital & Sachet Micro-Credit

PREDICTIVE DEFICIT HEDGING

Cashflow Velocity Radar

Maps upcoming 14-day obligations (supplier payables, rent, utility bills) against projected daily Soundbox inflow velocity.

- Identifies liquidity shortfalls 5 days before they occur.
- Calculates real-time net disposable cash position.
- Eliminates bounced cheques and penalty charges with distributors.

PAYTM SACHET 2.0

Autonomous Micro-Accrual Sweeps

Instead of painful lump-sum debt repayments, Qora implements continuous painless micro-deductions.

- Underwrites pre-approved credit (₹25,000 to ₹75,000) using Soundbox transaction consistency, requiring zero paperwork.
- Daily Balance Recovery (DBR): sweeps tiny 3%–5% micro-accruals from settlement batches until cleared.
- Average loan duration: 14–21 days with 98.4% automated recovery.

RELATIONSHIP-AWARE AI

Conversational Voice Khata Recovery

Solves the social discomfort of collecting informal customer credit without burning community bridges.

- Cogneer analyzes customer history, payment punctuality, and visit frequency.
- Sarvam Bulbul synthesizes polite, dialect-specific voice calls or WhatsApp voice notes in the customer's native language.
- Offers dynamic micro-installment payment links if full settlement is constrained.

1-Click GSTR-3B & Formal Credit Passport

Automatically categorizes Soundbox revenues and scanned supplier bills into GST tax brackets (0%, 5%, 12%, 18%). Generates an audited digital balance sheet that unlocks formal bank loans up to ₹5,00,000.

ZERO-PAPERWORK UNDERWRITING INSTANT DISBURSAL TO PAYTM WALLET

Compelling Economics for Both Merchant & Paytm

₹2,940 / mo

Direct Merchant ROI

Measured monthly value generated per Kirana store:

- ₹1,450 recovered from supplier overcharge claims
- ₹890 recovered in previously written-off khata debt
- ₹600 saved in prevented perishable stock write-offs
- ROI on Paytm device rental exceeds 20x from month 1.**

₹0 Rental

Sponsored Audio Co-Op Network

Converts the Soundbox from a cost center into a revenue generator:

- FMCG brands (HUL, ITC, Britannia) sponsor 2-second audio tags post-payment: *"Paytm par ₹150 praapt hue. Tide par paayein ₹5 discount."*
- Merchant earns ₹0.30 per sponsored playback.
- 15 daily ads = ₹135/mo, **offsetting Soundbox rental completely.**

1.2% / mo

Paytm Lending Origination

Paytm's high-margin monetization engine:

- Paytm earns origination & distribution fees from lending partners (Tata Capital, SMFG, Piramal).
- Near-zero default risk via programmatic daily DBR settlement sweeps.
- Drastically increases merchant stickiness and Soundbox retention.

Unassailable Strategic Moat

Competitors can copy payment alerts, but they cannot replicate Qora's triad: Cognee's multi-year merchant memory graph + Sarvam's sovereign Indic speech models + Paytm's 10M+ hardware distribution footprint.

Zero Hardware Capex

Qora operates completely on existing deployed Paytm Soundbox models (4G/Wi-Fi with speaker/mic) and WhatsApp. No new hardware rollout required.

From Working Prototype to 10M Merchants

A phased, production-grade roadmap designed for seamless integration with Paytm's infrastructure.

PHASE 1: DAYS 1 – 30

Bengaluru Pilot Cohort

Deploy WhatsApp-based Qora companion across 500 Kirana stores in Koramangala & HSR Layout.

- Validate Sarvam Saaras v4 accuracy on Kannada/Hindi code-mixing.
- Calibrate Cognee knowledge graphs on distributor invoices.

PHASE 2: DAYS 31 – 60

Soundbox Firmware Ingestion

Direct integration with Paytm AI Soundbox firmware via WebSocket APIs.

- Enable real-time voice queries directly on the counter device.
- Initiate automated DBR micro-accrual sweeps for sachet loans.

PHASE 3: DAYS 61 – 90

Nationwide Commercial Rollout

Roll out to top 10 retail hubs across Tier 1, 2, and 3 cities.

- Launch FMCG sponsored audio ad co-op network.
- Activate Federated Regional Price Oracle across major wholesale mandis.

Explore the fully functional dashboard, voice copilot, invoice auditor & cashflow engine.

fursatiinsaan.github.io/Qora github.com/fursatiinsaan/Qora